SNOWDON TOWERS ELECTRICAL VDC DRAWING PACKAGE

ST-VDC-01 | Isaak Meakin
Parking P05 / Level 1 Retail 109
Independent educational capstone
Autodesk Snowdon Towers sample-model context

NOT FOR CONSTRUCTION

MODELED SCOPE

Parking P05: two panels, one transformer, one disconnect and associated feeder pathways.

Level 1 Retail 109: six receptacles, J1, three parallel 2" EMT rack conduits, representative supports and a 12" x 4" cable tray.

DRAWING PURPOSE

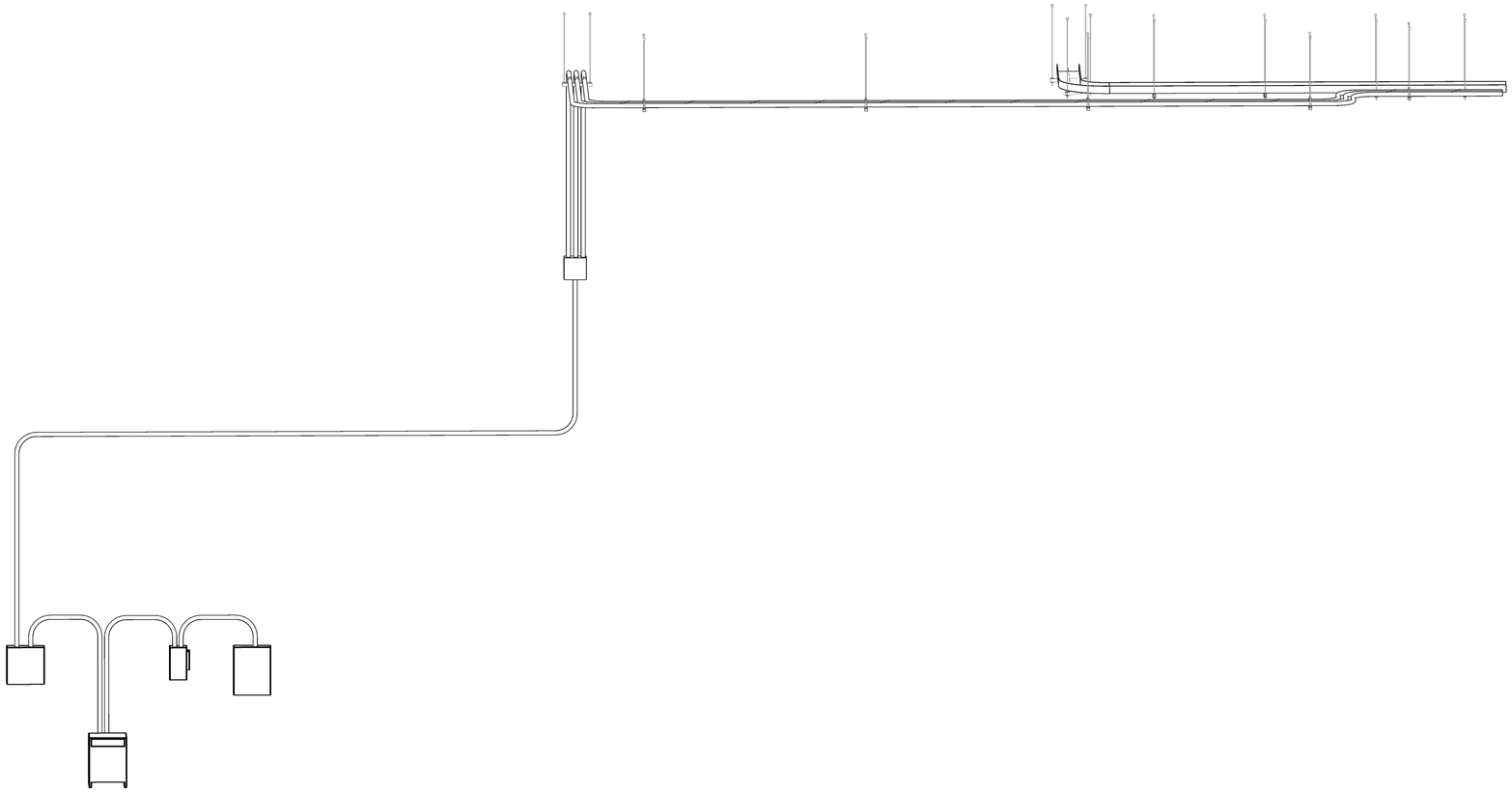
Document equipment placement, raceway layout, coordination relationships, hanger components and planned rack installation.

MODEL BASIS AND LIMITS

Original electrical working model by Isaak Meakin, using linked Autodesk sample models for context.

Routes beyond the modeled boundary are assumed continuations. Equipment ratings, support capacity, anchors and final service clearances are unverified.

Independent educational project — not for construction.



1 3D - Project Overview

Sheet Index	
Sheet Number	Sheet Name
E0.01	Cover & Sheet Index
E101	Retail 109 - Raceways
E102	Electrical Room - Equipment
E103	Raceway Coordination
IS1.01	Retail 109 - Rack Installation
MTO-01	Material Takeoff
PF1.01	Conduit Rack - Prefab Build

www.autodesk.com/revit

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

No.	Description	Date

Isaak Meakin

ST Electrical

Cover & Sheet Index

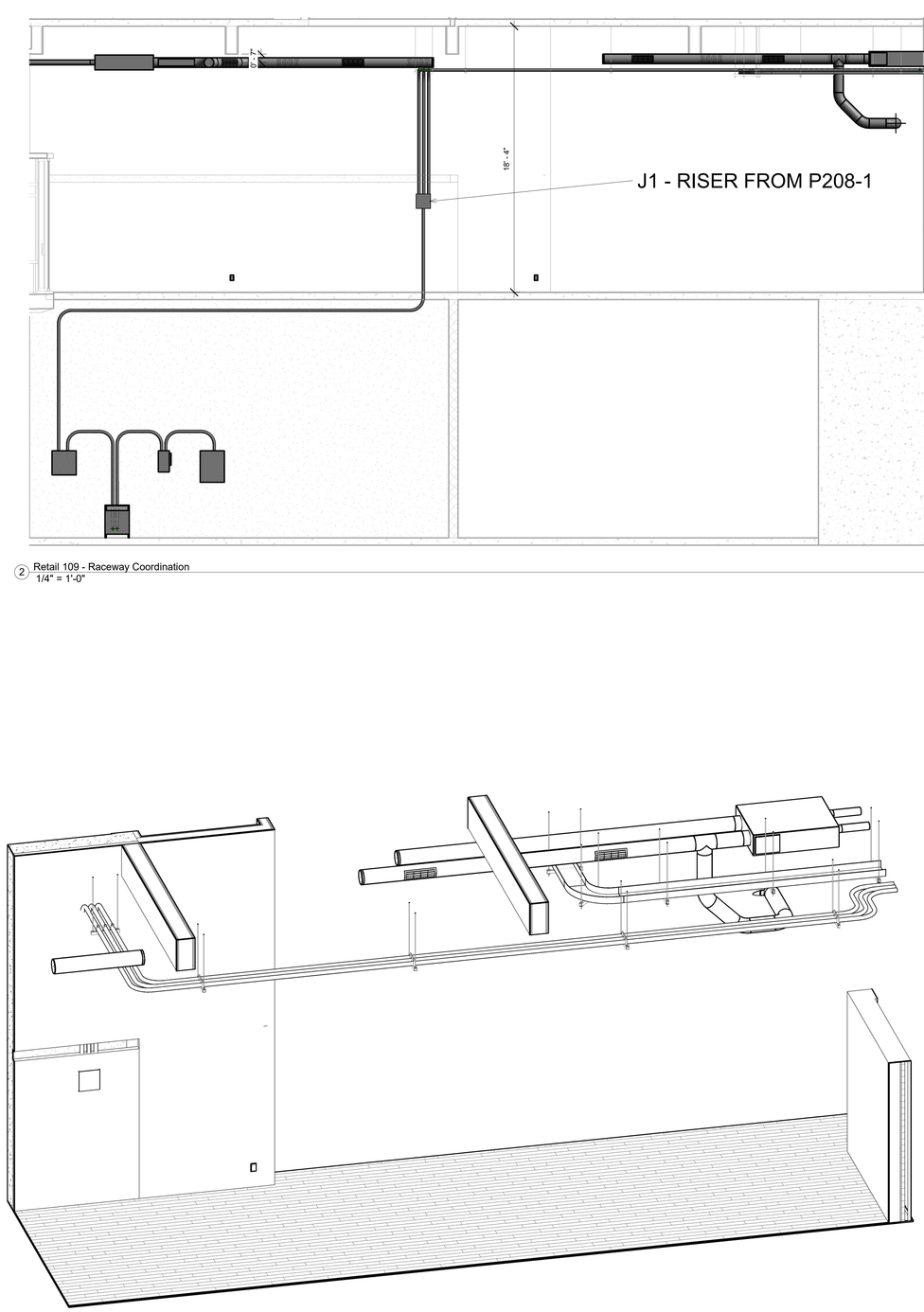
Project Number	ST-001
Date	10/8/26
Drawn By	Isaak Meakin
Checked By	Checker

E0.01

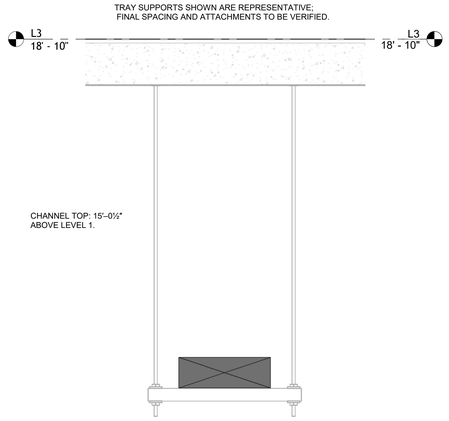
Scale

10/14/2026 3:15:10 PM

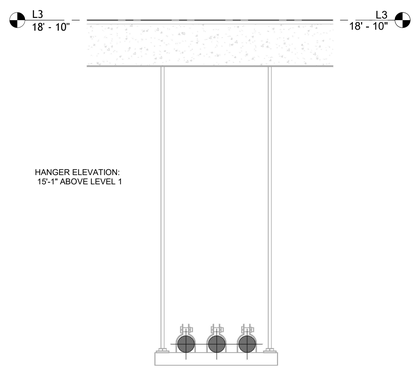
2 Retail 109 - Raceway Coordination
1/4" = 1'-0"



3 3D - L1 - Retail 109

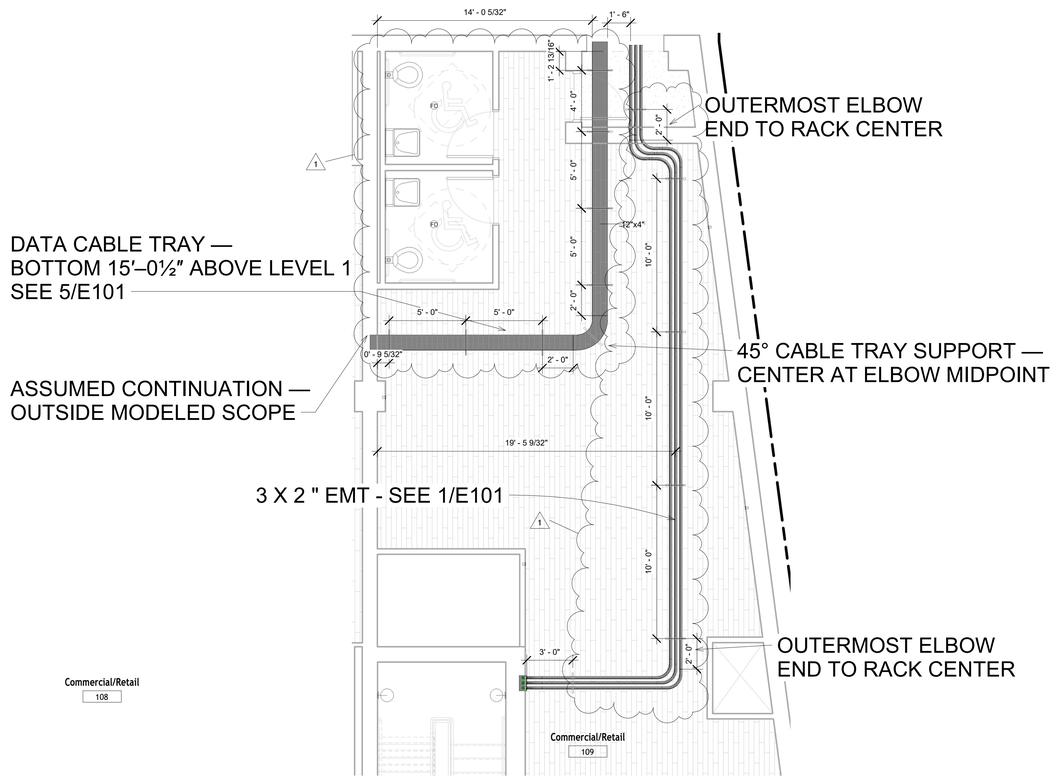


5 Retail 109 - Typical Cable Tray Detail
1 1/2" = 1'-0"



1 Retail 109 - Typical Conduit Rack Detail
1 1/2" = 1'-0"

CONDUIT & CABLE TRAY CONTINUE BEYOND MODEL SCOPE.
SUPPORTS BEYOND THIS POINT NOT SHOWN



4 Retail 109 - Raceway Plan
1/4" = 1'-0"

www.autodesk.com/revit

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

No.	Description	Date
1	FC-01-Raceway support layout and quantities revised	10/14/26

Isaak Meakin

ST Electrical

Retail 109 - Raceways

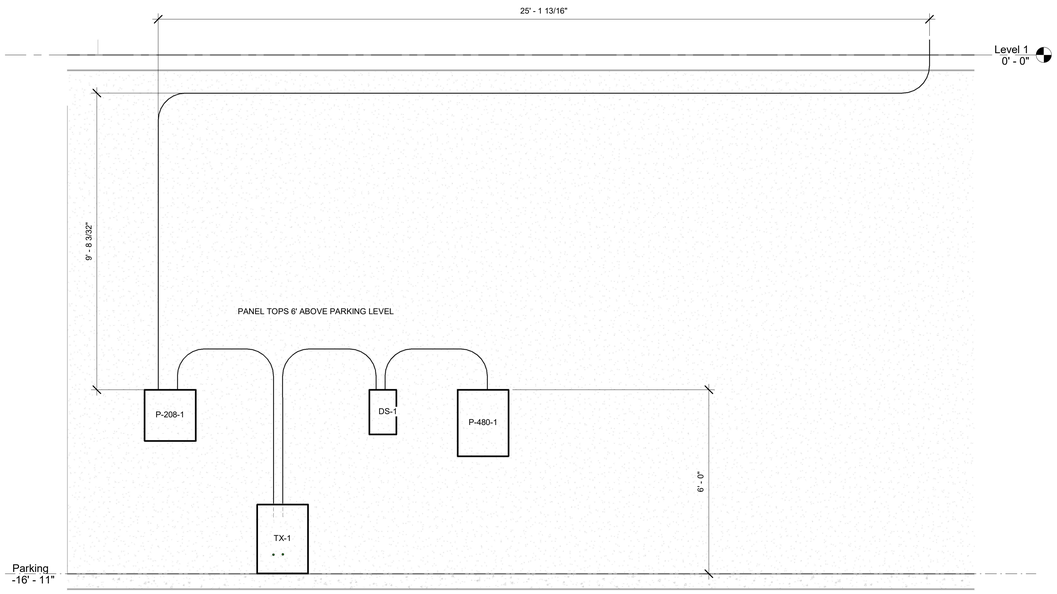
Project Number	ST-001
Date	10/8/26
Drawn By	Isaak Meakin
Checked By	Checker

E101

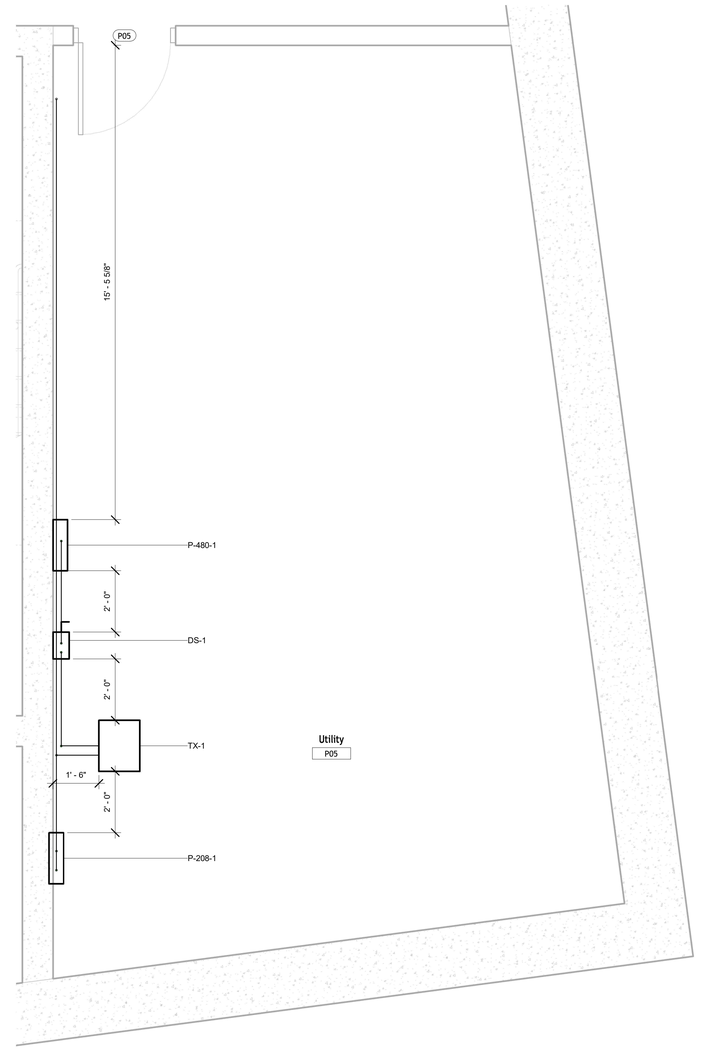
Scale	As indicated

10/14/2026 3:15:15 PM

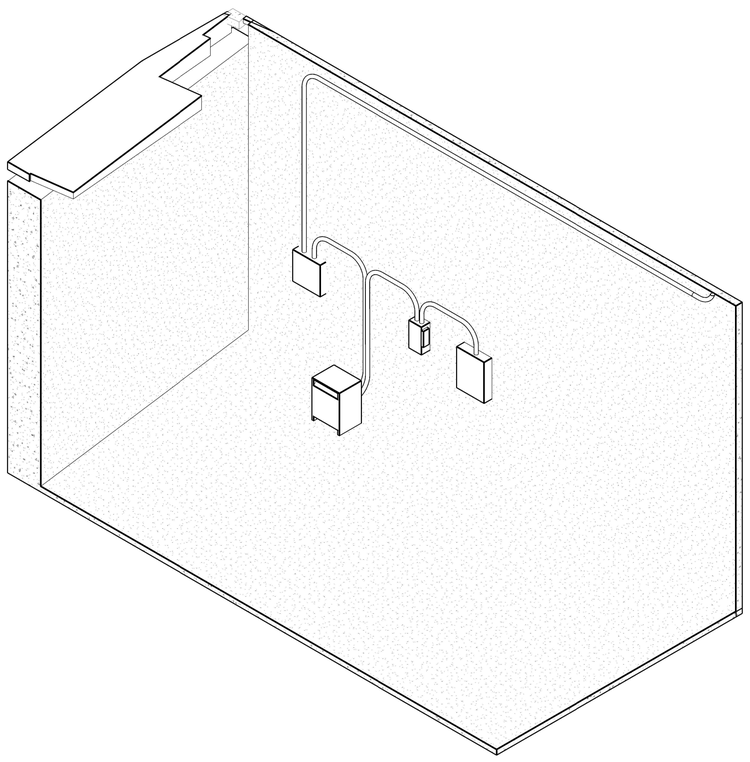
INDEPENDENT EDUCATIONAL PROJECT — NOT FOR CONSTRUCTION



2 P05 - Equipment Wall Elevation
1/2" = 1'-0"



1 Parking P05 - Electrical Room Plan
1/2" = 1'-0"



3 3D - Parking - P05

www.autodesk.com/revit

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

No.	Description	Date

Isaak Meakin

ST Electrical

Electrical Room - Equipment

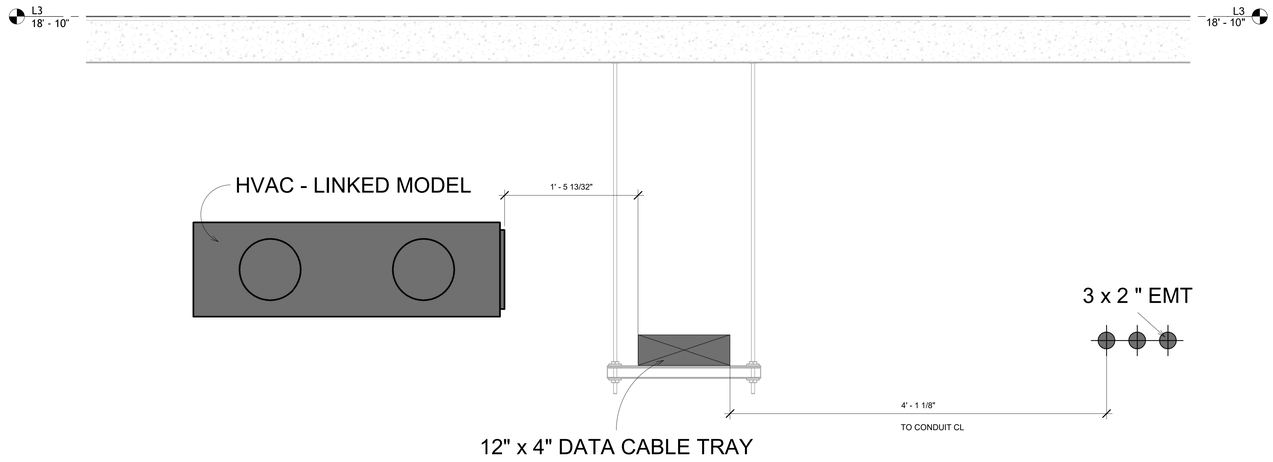
Project Number	ST-001
Date	10/8/26
Drawn By	Isaak Meakin
Checked By	Checker

E102

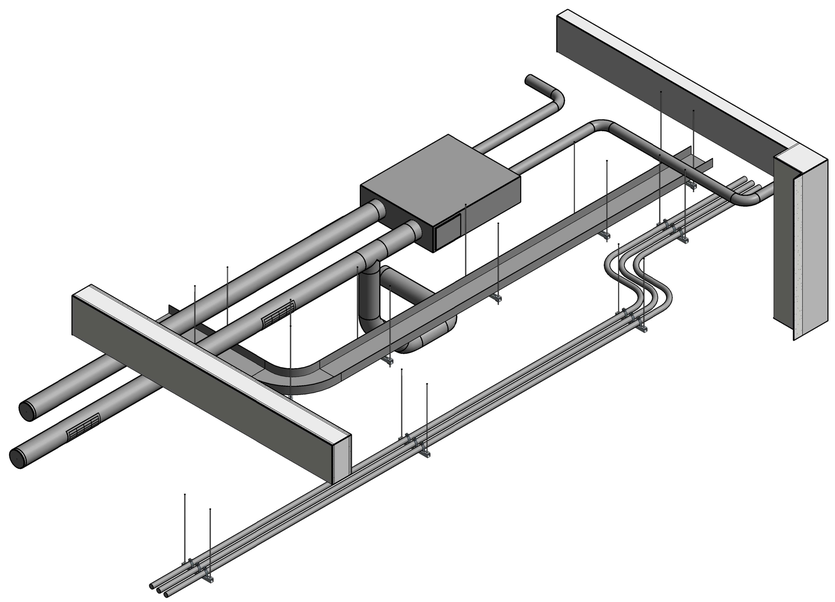
Scale	1/2" = 1'-0"

10/14/2026 3:15:19 PM

INDEPENDENT EDUCATIONAL PROJECT — NOT FOR CONSTRUCTION



1 Retail 109 – HVAC & Raceway Coordination
Section
1 1/2" = 1'-0"



2 3D - L1 - Retail 109 Raceway Coordination

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

www.autodesk.com/revit

No.	Description	Date
1	FC-01-Raceway support layout and quantities revised	10/14/26

Isaak Meakin

ST Electrical

Raceway Coordination

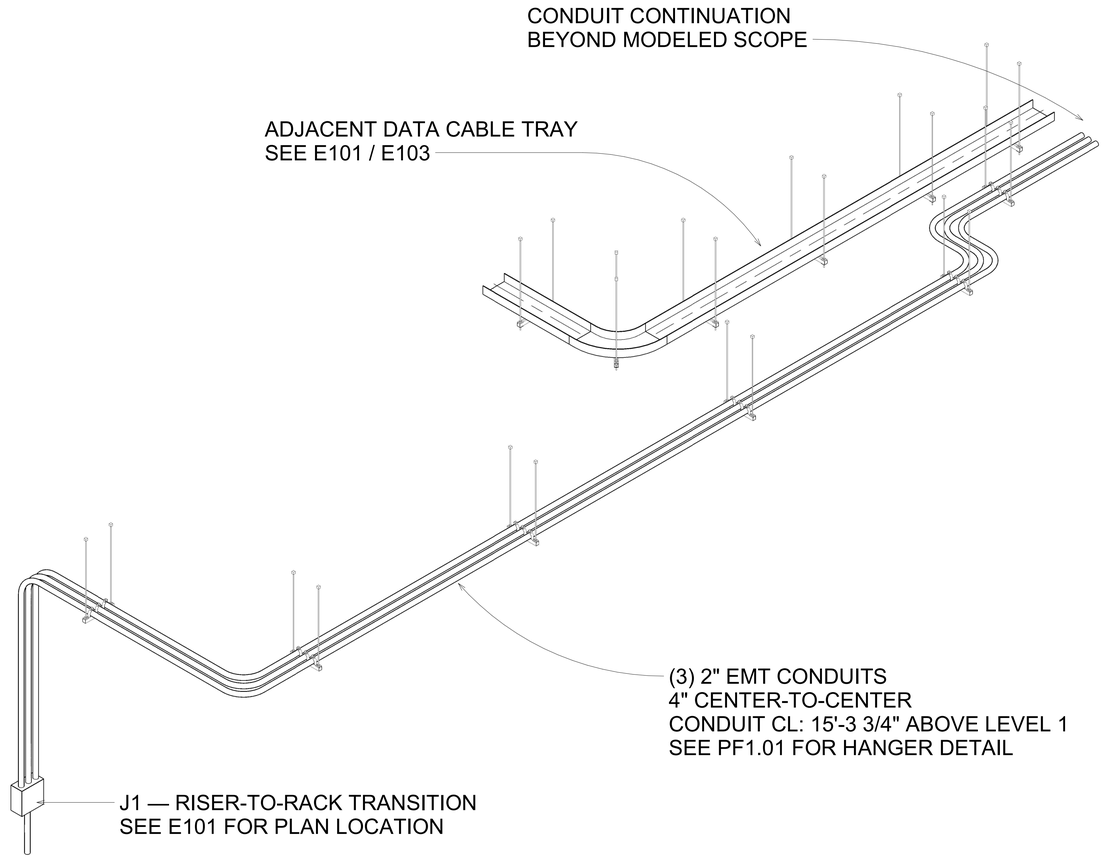
Project Number	ST-001
Date	10/8/26
Drawn By	Isaak Meakin
Checked By	Checker

E103

Scale 1 1/2" = 1'-0"

10/14/2026 3:15:23 PM

INDEPENDENT EDUCATIONAL PROJECT — NOT FOR CONSTRUCTION



1 3D - Retail 109 Rack Installation

RACK INSTALLATION — PLANNED SEQUENCE

1. Verify site dimensions, access and approved structural attachment details before installation.

2. Lay out the rack route and support locations using the dimensioned plan. See E101 for location and E103 for coordination sections.

3. Assemble the hanger components shown on PF1.01. Confirm support spacing and anchor requirements before placing supports.

4. Position the three 2" EMT conduits at 4" center-to-center spacing. Align the rack and verify its installation elevation against the section.

5. Complete the modeled J1 riser-to-rack transition. Check the bends and maintain the coordinated relationship with the adjacent cable tray and HVAC.

6. Check alignment, connections and clearances. Conduit ends at the modeled limit represent continuation beyond this scope.

REFERENCES
E101 — Raceway layout
E103 — Raceway coordination
PF1.01 — Rack detail and hanger BOM

Independent educational project — not for construction. Support capacity and structural attachments are unverified. Do not scale the 3D view.

www.autodesk.com/revit

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

No.	Description	Date
1	FC-01-Raceway support layout and quantities revised	10/14/26

Isaak Meakin

ST Electrical

Retail 109 - Rack Installation

Project Number	ST-001
Date	10/8/26
Drawn By	Isaak Meakin
Checked By	Checker

IS1.01

Scale

10/14/2026 3:15:25 PM

MTO-01 - Electrical Equipment

Mark	Family	Type	Panel Name	Level	Count
DS-1	Disconnect Switches	480V - Fusible 60A		Parking	1
P-208-1	Lighting and Appliance Panelboard - 208V MLO - Surface	100 A	P-208-1	Parking	1
P-480-1	Lighting and Appliance Panelboard - 480V MLO - Surface	125 A	P-480-1	Parking	1
TX-1	Dry Type Transformer - 480-208Y120 - NEMA Type 2	30 kVA	TX-1	Parking	1

Grand total: 4

MTO-01-Cable Tray Fittings

Family and Type	Size	Count
Channel Horizontal Bend: Standard	12"x4"-12"x4"	1

Grand total: 1

MTO-01-Conduit Fittings

Family and Type	Size	Count
Conduit Elbow - without Fittings - EMT: Standard	2"ø-2"ø	22
Conduit Junction Box - Tee - Aluminum: Standard	2"ø-2"ø-2"ø-2"ø	1

Grand total: 23

MTO-01-Supports

Family and Type	Count	Comments
Atkore Cable Tray Hanger Fri Oct 09 2026: Atkore Cable Tray Hanger Fri Oct 09 2026	8	Cable tray trapeze - 20″ channel
Atkore Pipe Conduit Hanger Wed Oct 07 2026(1): Atkore Pipe Conduit Hanger Wed Oct 07 2026	6	Conduit rack trapeze - 16″ channel

Grand total: 14

MTO-01-Conduit Takeoff

Family and Type	Diameter(Trade Size)	Length	Count
Conduit without Fittings: Electrical Metallic Tubing (EMT)	2"	207' - 1 25/32"	29
Grand total: 29		207' - 1 25/32"	

MTO-01-Cable Tray Takeoff

Family and Type	Width	Height	Length	Count
Cable Tray with Fittings: Wire Mesh Cable Tray	12"	4"	31' - 1 3/8"	2
Grand total: 2			31' - 1 3/8"	

TAKEOFF NOTES
1. Quantities cover the host electrical model only; linked models excluded.
2. Lengths cover modeled segments; fitting lengths, waste and stock allowances excluded.
3. Supports counted as assemblies. See PF1.01 for conduit-rack component BOM.
4. Wire, anchors and unmodeled connection hardware excluded.
5. Independent educational project; not for construction or purchasing.

AUTODESK

www.autodesk.com/revit

Isaak Meakin

ST Electrical

No.	Description	Date
1	FC-01-Raceway support layout and quantities revised	10/14/26

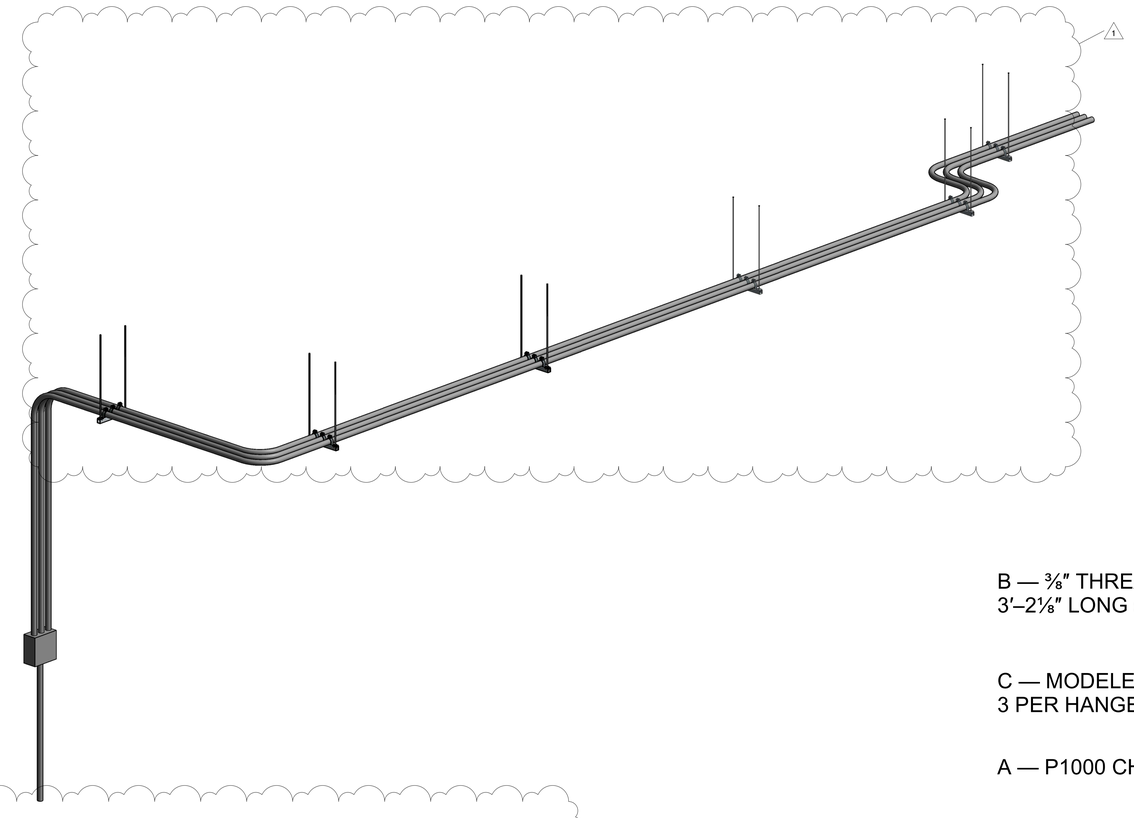
Material Takeoff

Project number	ST-001
Date	10/8/26
Drawn by	Isaak Meakin
Checked by	Checker

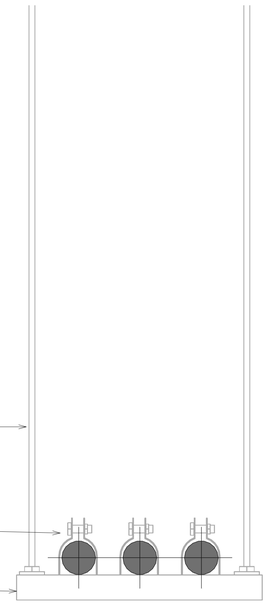
MTO-01

Scale

10/14/2026 3:04:42 PM



1 3D - Prefab Conduit Rack



2 SEC - Prefab Conduit Rack
3" = 1'-0"

HANGER COMPONENT BOM — 6 MODELED HANGERS

P1000 CHANNEL — 16" LONG — QTY: 6 EA
THREADED ROD — 3/8" DIA × 3'-2 1/8" LONG — QTY: 12 EA
MODELED 2" CLAMP — P1117 — QTY: 18 EA

Quantities derived from six matching modeled hangers.
Connection hardware and anchors are not included.
Conduit lengths are documented in the material takeoff.
Educational prefab detail — not for construction.

www.autodesk.com/revit

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

Consultant
Address
Address
Address
Phone

No.	Description	Date
1	FC-01-Raceway support layout and quantities revised	10/14/26

Isaak Meakin

ST Electrical

Conduit Rack - Prefab Build

Project Number	ST-001
Date	10/8/26
Drawn By	Isaak Meakin
Checked By	Checker

PF1.01

Scale 3" = 1'-0"

10/14/2026 3:15:28 PM